

RESEARCH

How do humans use acoustic-prosodic cues to facilitate conversation? How can computational systems leverage this information?

- Analysis of discourse structure behaviors, and their interaction with stance and surprisal
- Speech recognition performance as a function of human confidence and prosodic variation.

EDUCATION

University of Washington

Ph.D. in Linguistics, 2024

M.S. in Computational Linguistics, 2023

Advised by: Richard Wright,
Mari Ostendorf

Thesis: *Prosody in Human Communication and Machine Understanding*

University of Utah

Honors B.A. Linguistics, 2017

B.S. Applied Mathematics, 2017

TESOL Certificate, 2016

Advised by: Abby Kaplan

Thesis: *Musical Text-Setting as Evidence for Syllabification of Highly Moraic Structures in English*

SKILLS

Languages:

Python, R, Matlab, HTML/CSS, JS

Machine Learning:

pytorch, tensorflow, Scikit-learn

Linguistic Tools:

Praat, Kaldi, NLTK

Natural Languages:

French (basic)

Spanish (basic)

Toisan Chinese (heritage)

Specialized Equipment:

electroglottograph, field recorder, tongue ultrasound

WORK EXPERIENCE

Simon Fraser University

July 2025 - present

Assistant Professor

- Joint appointee in Linguistics and Social Data Analytics

Western Washington University

Sept 2024 - June 2025

Visiting Assistant Professor

- Teaching: Intro. to Phonetics, Computational Linguistics, Honors Colloq. in Linguistics, Language & Society, Corpus Linguistics, Patterns in Your Mind
- Service: Computational linguistics curriculum committee member

University of Washington

Sept 2017 – June 2024

Teaching and Research Assistant

- Teaching: Intro. to Phonetics, Computational Linguistics, Shallow Processing Techniques for NLP, Swearing & Taboo Language
- Research assistant for computational modeling and language documentation projects

Meta

Research Scientist Intern, Reality Labs

June 2022 - Dec. 2022
& Summer 2023

- Developed classifier of hearing impairment from behavioral cues for use in Augmented Reality

Google

SWE Intern, Federated Assistant

June 2021 - March 2022

- Identified causes of failure for existing data augmentation strategies
- Computational Linguist Intern, Pygmalion Summer 2019
- Designed prosodic DNN to classify code-switching

RECENT PUBLICATIONS

Probing Whisper Predictions for French, English and Persian Transcriptions

2024; Nicolas Ballier, Léa Burin, Behnoosh Namdarzadeh, Sara Ng, Richard Wright, Jean-Baptiste Yunès; ICNLSP

Investigating the Influence of Stance-Taking on Conversational Timing of Task-Oriented Speech

2024; Sara Ng, Gina-Anne Levow, Mari Ostendorf, Richard Wright; INTERSPEECH

Effects of Information Load and Pragmatic Load on the Hypo-Hyper Continuum

2024; Sara Ng, Valerie Freeman, Gina-Anne Levow, Mari Ostendorf, Richard Wright; Fonetik

Investigating the Acoustic Fidelity of Vowels across Remote Recording Methods

2024; Marina Oganyan, Gina-Anne Levow, Robert Squizzero, Emily P. Ahn, Sara Ng, Ella Deaton, Richard A. Wright; Ling. Vanguard

Properties of Constructed Language Phonological Inventories

2023; Sara Ng, Abigail Schwendiman Sleight; UWWPL

Leveraging Prosody for Punctuation Prediction of Spontaneous Speech

2022; Yeonjin Cho, Sara Ng, Trang Tran, Mari Ostendorf; INTERSPEECH